Euclid's Elements — Book XI
Proposition 1

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJteamLab

A part of a straight line cannot be in the plane of reference and a part in plane more elevated.

1 Introduction

Proposition XI.1 opens Euclid's solid geometry with a statement whose mathematical content is fundamentally incidence-theoretic. No angle, congruence, order, parallelism, or continuity is involved. The intended content is that once two distinct points of a straight line lie in a plane, the whole straight line lies in that plane.

This proposition is therefore an especially useful comparison point between Euclid's organization of Book XI and Hilbert's axiomatic organization of space. In the present formal reconstruction the proposition is not proved by reproducing Euclid's historical circle argument. Its precise incidence content is already one of the basic spatial incidence principles: Hilbert I.6.

The production file is

```
CGJteamLab/Proposition11_1.lean
```

and it imports only

```
CGJteamLab.Hilbert3DInterface
```

from the project-level spatial infrastructure.

2 Euclid's Statement

Euclid states Proposition XI.1 as follows:

> A part of a straight line cannot be in the plane of reference and a part in plane more elevated.

The natural precise incidence reading is:

> If two distinct points of a straight line lie in a plane, then the whole straight line lies in that plane.

This is the interpretation used by the formal reconstruction.

3 Classical Configuration

Let l be a straight line and let π be the reference plane. Choose two distinct points A, B on l such that

$$A \in \pi, \qquad B \in \pi.$$

The formal target is

$$l \subset \pi.$$

Euclid describes the same situation diagrammatically by regarding a segment AB as lying in the reference plane and supposing, for contradiction, that a continuation of the same straight line leaves that plane.

There is no essential metric configuration. The proposition concerns only the relation between a line, two of its points, and a plane.

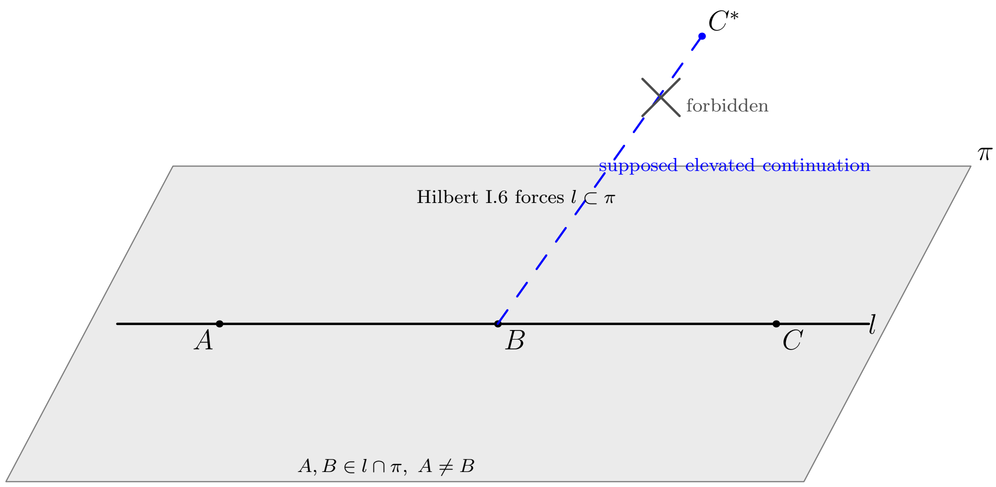

Figure 1: The incidence content of Proposition XI.1. The black carrier l contains the distinct points A, B of the reference plane π, hence Hilbert I.6 forces the whole line to remain in π. The blue dashed segment toward C^* does not represent a realizable straight continuation; it records only Euclid's contradictory supposition that the same straight line could leave the plane.

4 Euclid's Classical Proof

The received proof proceeds by contradiction. Euclid supposes that a part AB of a straight line ABC lies in the reference plane while the continuation BC is elevated out of that plane.

He then chooses in the reference plane another straight continuation BD of AB. The lines ABC and ABD would therefore share the segment AB. Euclid declares this impossible and appeals to a circle with center B and radius AB.

The logical difficulty is that this circle argument is not an adequate spatial incidence foundation. In three-dimensional space the center and radius do not determine a unique circle unless its plane is also specified. More importantly, the desired conclusion is itself an elementary closure property of line–plane incidence and should not require metric circle geometry.

For the formal reconstruction we therefore retain the mathematical incidence content of XI.1 but not this historical argument.

5 Classical Dependency Pattern

At the level of the printed proof, the structure is roughly

$$\begin{gathered}
AB \text{ lies in the reference plane} \\
\Downarrow \\
\text{suppose the continuation } BC \text{ leaves the plane} \\
\Downarrow \\
\text{construct another continuation } BD \text{ in the plane} \\
\Downarrow \\
ABC \text{ and } ABD \text{ share } AB \\
\Downarrow \\
\text{circle-based contradiction} \\
\Downarrow \\
\text{the straight line cannot leave the plane.}
\end{gathered}$$

This is not the dependency graph used by the formal proof.

6 What Is Genuinely Three-Dimensional Here?

XI.1 is spatial only because the primitive object π is an ambient plane and the conclusion concerns containment of an ambient line in that plane. There is no change of plane and no interaction between two distinct planes.

The genuine three-dimensional datum is simply

$$A, B \in l \cap \pi, \qquad A \neq B.$$

From these data the conclusion is the closure statement

$$l \subset \pi.$$

Thus XI.1 belongs entirely to the spatial incidence layer. It does not require an induced planar geometry PlaneGeo(π), because there is no planar theorem to prove inside π.

7 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_1
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi : S.Plane)
    (l : Geo.Line)
    (A B : Geo.Point)
    (hAB : Ne A B)
    (hAl : H.OnLine A l)
    (hBl : H.OnLine B l)
    (hApi : S.OnPlane A pi)
    (hBpi : S.OnPlane B pi) :
    HilbertLineInPlane Geo l pi := by
  exact
    HilbertSpaceIncidence.line_in_plane
      (Geo := Geo)
      A B hAB
      l hAl hBl
      pi hApi hBpi
```

The conclusion

```
HilbertLineInPlane Geo l pi
```

means that every point incident with l is incident with π.

8 Hilbert Incidence Architecture

The decisive declaration is the spatial incidence field

```
HilbertSpaceIncidence.line_in_plane :
  forall A B : Geo.Point,
    Ne A B ->
    forall l : Geo.Line,
      H.OnLine A l ->
      H.OnLine B l ->
      forall pi : S.Plane,
        S.OnPlane A pi ->
        S.OnPlane B pi ->
        HilbertLineInPlane Geo l pi
```

This is Hilbert incidence axiom I.6 in the current 3D organization:

> If two distinct points of a line lie in a plane, then every point of the line lies in that plane.

Consequently the formal Euclid theorem is intentionally a thin proposition wrapper over the general spatial incidence interface.

This is structurally analogous to several Book I reconstruction points where a Euclid proposition becomes short after its reusable mathematical content has already been isolated in a general Hilbert theorem or axiom interface.

9 Ambient Space and PlaneGeo

No `PlaneGeo` slice is created in XI.1.

The objects remain ambient throughout:

```
A B : Geo.Point
l   : Geo.Line
pi  : S.Plane
```

and the incidence relations are

```
H.OnLine A l
H.OnLine B l
S.OnPlane A pi
S.OnPlane B pi
```

The proof does not require

```
HilbertPlaneIncidence Geo
HilbertSpaceOrder Geo
HilbertSpaceCongruence Geo
HilbertSpaceEuclidean Geo
```

as theorem parameters.

This is the minimal spatial layer appropriate to the proposition.

10 Lean Proof Architecture

The Lean proof has exactly one mathematical step:

$$
\begin{gathered}
A \neq B \\
A \in l, \quad B \in l \\
A \in \pi, \quad B \in \pi \\
\Downarrow \\
\texttt{HilbertSpaceIncidence.line\_in\_plane} \\
\Downarrow \\
l \subset \pi.
\end{gathered}
$$

There are no proposition-local helper lemmas and no case distinctions.

The formal proof does not construct a second line, a circle, an auxiliary plane, or any metric object. The historical contradiction is replaced by the exact spatial incidence principle that expresses the intended content.

11 Hidden Formal Data

Although the theorem is short, two pieces of data that are implicit in Euclid's prose are explicit in Lean.

11.1 The two plane points must be distinct

The hypothesis

```
hAB : Ne A B
```

is essential. A single point shared by a line and a plane does not force the whole line to lie in that plane.

11.2 The line carrier is explicit

Euclid speaks of the straight line ABC. Lean receives an explicit carrier

```
l : Geo.Line
```

together with the incidence witnesses

```
hAl : H.OnLine A l
hBl : H.OnLine B l
```

The conclusion is not merely that a third named point lies in the plane; it is the universal carrier statement `HilbertLineInPlane Geo l pi`.

12 Relationship with Earlier and Later Book XI Propositions

XI.1 has no earlier Book XI proposition dependency.

Its significance is forward-looking. The closure principle expressed by XI.1 is repeatedly needed whenever a later spatial argument knows two points of a line to lie in a plane and must promote that pointwise information to whole-line containment.

In the present Hilbert architecture later propositions normally use the general declaration

```
HilbertSpaceIncidence.line_in_plane
```

directly rather than importing `euclid_proposition_11_1`. Thus the mathematical content of XI.1 is foundational, while the Euclid-numbered theorem records the correspondence between the *Elements* and the general 3D incidence layer.

XI.2 is the first immediate classical consumer of XI.1: Euclid uses the principle that a line determined inside a plane cannot subsequently leave that plane. In the formal development XI.2 can instead call the stronger general spatial theorem for two intersecting lines.

13 Relationship with Hilbert's Group I

XI.1 is a particularly direct meeting point between Euclid and Hilbert.

In the current project, Hilbert's spatial incidence layer contains:

- existence of points on lines;
- existence and uniqueness of planes through noncollinear points;
- line closure under two plane points;
- the common-point principle for two planes;
- existence of noncoplanar points.

XI.1 corresponds exactly to the line-closure clause, Hilbert I.6.

No order axiom from Group II is used. No congruence axiom from Group III is used. No Euclidean parallel axiom from Group IV is used. No continuity assumption is used.

14 Classical DAG versus Formal DAG

The central structural difference can be summarized as follows.

Euclid.

$$\begin{gathered}\text{supposed line leaving the reference plane}\\ \Downarrow\\ \text{second continuation in the reference plane}\\ \Downarrow\\ \text{shared segment}\\ \Downarrow\\ \text{circle-based contradiction}\\ \Downarrow\\ \text{XI.1.}\end{gathered}$$

Lean / Hilbert.

$$\begin{gathered} A \neq B, \quad A, B \in l, \quad A, B \in \pi \\ \Downarrow \\ \text{Hilbert I.6} \\ \Downarrow \\ \texttt{HilbertSpaceIncidence.line\_in\_plane} \\ \Downarrow \\ \texttt{euclid\_proposition\_11\_1}. \end{gathered}$$

The second DAG is shorter because the incidence principle needed to make solid geometry possible is stated explicitly at the axiomatic level.

15 Structural Lesson

XI.1 exposes a foundational issue that is easy to miss when reading Book XI diagrammatically. To begin solid geometry one needs rules governing how lines and planes are incident. A planar circle construction cannot replace those rules.

The Hilbert reconstruction makes this boundary explicit. The proposition contains no metric content to reconstruct: its correct mathematical home is the spatial incidence system itself.

This also explains why the production theorem is deliberately thin. The formalization is not avoiding Euclid's mathematical content; it is locating that content at the level where it functions as a foundational rule for all later spatial arguments.

16 Audit Status

This documentation was checked against the current production declaration `Geometry.euclid_proposition_11_` and the spatial incidence interface used by its proof. The formal statement, the Hilbert incidence interpretation, and the classical/formal dependency comparison in this note reflect the current production architecture.

17 Conclusion

Proposition XI.1 is the incidence threshold of Book XI. Its intended mathematical statement is simple:

$$A \neq B, \quad A, B \in l, \quad A, B \in \pi \quad \Longrightarrow \quad l \subset \pi.$$

In the current formal architecture this is exactly Hilbert I.6. The Lean theorem therefore contains no auxiliary construction and no proposition-local proof machinery: it exposes the already explicit spatial incidence rule under Euclid's proposition number.

The contrast with the historical proof is instructive. Euclid attempts to exclude a line leaving the plane by a circle-based contradiction. The Hilbert reconstruction instead identifies the missing foundational content directly as line–plane incidence closure. That is both logically cleaner and exactly the form required by the subsequent development of solid geometry.